Macroeconomics

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1 Fundamental Concepts

Economics is the social science that studies the production, distribution, and consumption of goods and services. At its core, economics addresses the fundamental problem of **scarcity**: while resources such as labor, capital, and natural resources are limited, human wants are essentially unlimited. This creates the central challenge that all economies must navigate—how to allocate scarce resources among competing uses.

Core Concept 1.1: Factors of Production

The production of goods and services requires four essential factors of production:

(a) **Land** (*also known as Natural Resources*) encompasses all natural inputs used in production, including agricultural land, mineral deposits, water resources, and the physical space upon which production facilities are constructed.

(b) **Labor** represents the human effort devoted to production, encompassing both physical and mental work. Labor includes the services of workers across all sectors, from agricultural laborers harvesting crops to knowledge workers developing software.

(c) **Capital** refers to goods that are produced and then used to produce other goods and services. Unlike consumer goods that satisfy wants directly, capital goods facilitate future production. Examples include manufacturing facilities, machinery, transportation equipment, and tools.

(d) **Entrepreneurship** represents the creative effort of individuals who combine the other three factors of production to create goods and services. Entrepreneurs bear risk, organize production, and drive innovation.

Core Concept 1.2: Opportunity Cost

When making economic decisions, individuals and societies must choose among mutually exclusive alternatives. The **opportunity cost** of a choice represents the value of the best alternative forgone. This concept is fundamental to rational decision-making because choosing one option necessarily means sacrificing another.

Core Concept 1.3: Production Possibilities Curve

The **Production Possibilities Curve (PPC)** illustrates the maximum combinations of two goods that an economy can produce given its available resources and technology.

(a) Points **on the curve** represent efficient production where all resources are fully utilized.

(b) Points **inside the curve** indicate underallocation of resources

(c) Points **outside the curve**—although appearing unattainable—can occur temporarily during periods of economic overheating when resources are used beyond their sustainable capacity. However, this state is not sustainable and typically leads to inflation, resource depletion, or economic crisis.

< PLOT OF PRODUCTION POSSIBILTIES CURVE >

The **shape of the PPC** reveals important information about opportunity costs. A concave curve (FIGURE REFERENCE) indicates increasing opportunity costs, meaning that producing additional units of one good requires sacrificing increasingly larger quantities of the other good. This typically reflects the fact that resources are not equally adaptable to producing both goods.

Shifts in the PPC represent economic growth or decline. **Outward shifts indicate economic growth**, resulting

from increases in available resources, improvements in technology, or enhancements in human capital. On the other hand, **inward shifts indicate economic decline**, potentially from natural disasters, war, or resource depletion.

Core Concept 1.4: Comparative and Absolute Advantage

An economy has an **absolute advantage** when it can produce more of a good than another economy using the same resources. **Comparative advantage** is when an economy can produce a good at a lower opportunity cost than another economy.

It is impossible for one economy to have a comparative advantage in both goods simultaneously. This principle, known as the **law of comparative advantage**, implies that all economies can benefit from trade by specializing in goods where they hold a comparative advantage. The gains from trade arise from this specialization and the subsequent exchange of goods.

Core Concept 1.5: Demand

Demand represents the quantity of a good that consumers are willing and able to purchase at various prices during a given time period. The **Law of Demand** states an inverse relationship between price and quantity demanded: as prices rise, quantity demanded falls, and vice versa. This relationship reflects both the **substitution effect** (consumers shift to cheaper alternatives) and the **income effect** (higher prices reduce purchasing power).

< PLOT OF DEMAND GRAPH >

The determinants of demand, factors that can cause the entire demand curve to shift, include:

(a) **Consumer income**: Normal goods experience increased demand as income rises, while inferior goods experience decreased demand.

(b) **The number of buyers** in the market.

(c) **The prices of substitute and complementary goods**.

(d) **Consumer expectations** about future prices.

(e) **Consumer tastes and preferences**.

Core Concept 1.6: Supply

Supply represents the quantity of a good that producers are willing and able to offer for sale at various prices. The **Law of Supply** indicates a direct relationship between price and quantity supplied: higher prices incentivize producers to increase production.

< PLOT OF SUPPLY GRAPH >

Determinants of supply include:

(a) **Production costs**, including input prices and technology.

(b) **The prices of related goods** firms could produce.

(c) **Expectations about future prices**.

(d) **The number of sellers** in the market.

(e) **Government policies**, such as taxes, subsidies, and regulations.

Core Concept 1.7: Market Equilibrium

Market Equilibrium occurs where the supply and demand curves intersect. At this point, the quantity supplied equals the quantity demanded, and the market clears. The equilibrium price and quantity are determined automatically through the price mechanism without external intervention. Correspondingly, adjustments in equilibrium price occur when supply and demand curve shifts.

< PLOT OF MARKET EQUILIBRIUM >

Core Concept 1.8: Circular Flow of Income

The **Circular Flow of Income** illustrates how goods, services, and money flow between **households** and **firms** in an economy. Firms produce goods and services using factors of production provided by households and, in exchange, pay households wages, rent, interest, and profits. Households then use this income to purchase goods and services from firms, completing the circular flow.

< FIGURE OF THE CIRCULAR FLOW OF INCOME >

This model (FIGURE REFERENCE) illustrates the interdependence between sectors of the economy and forms the basis for measuring aggregate economic activity through Gross Domestic Product (CORE CONCEPT REFERENCE).

2 Economic Indicators

Macroeconomic analysis relies on various **economic indicators** to assess the health and performance of an economy. These indicators provide insight into economic trends, business cycles, and the effectiveness of policy interventions.

2.1 Gross Domestic Product

Core Concept 2.1: Gross Domestic Product

Gross Domestic Product (GDP) represents the total market value of all **final** goods and services produced within a country's borders during a specific time period, typically one year or one quarter. GDP does not include the following items.

(a) **Transfer payments**: Government payments not tied to current production, such as Social Security.

(b) **Intermediate goods**: Which are counted in final goods to avoid double-counting.

(c) **Financial transactions**, like stock purchases.

(d) **Goods produced in previous periods**, such as existing home sales.

In general, GDP serves as the primary measure of economic output and standard of living.

Core Concept 2.2: Economic Cycles

The **business cycle** represents fluctuations in economic activity around the long-term growth trend. The cycle comprises four phases: expansion (real GDP rising), peak (the highest point of economic activity), recession

(real GDP falling), and trough (the lowest point).

< PLOT OF ACTUAL GDP TO POTENTIAL GDP >

As shown in FIGURE REFERENCE, the **output gap** measures the difference between actual GDP and potential GDP (*the level of GDP that would exist if all resources were fully utilized*). A positive output gap indicates the economy is operating above its sustainable capacity, while a negative output gap indicates underutilization of resources.

Economists typically define a **recession** as two consecutive quarters of negative real GDP growth, though the National Bureau of Economic Research uses more comprehensive criteria.

GDP can be calculated through multiple approach that would yield equivalent results

Core Concept 2.3: Calculating GDP (*Expenditures Approach*)

The **expenditures approach**—the most common approach—sums all spending on final goods and services.

$$\mathbf{GDP = C + I + G + X - M}$$

Here,

(a) **C** represents household consumption (*spending on final goods and services*).

(b) $\mathbf{I}_g$ represents gross investment (*spending on capital goods and inventory, excluding intermediate goods*).

(c) **G** represents government purchases (*government spending on goods and services, excluding transfer payments*).

(d) **X** represents exports.

(e) **M** represents imports.

Core Concept 2.4: Net Domestic Product (NDP)

Net domestic product (NDP) is GDP but accounts for the depreciation of capital goods. Its calculation takes the same method as the expenditures approach (CORE CONCEPT REFERENCE), but substitutes gross investment with net investment. **Net investment ($\mathbf{I}_n$)** is gross investment minus depreciation (*the decrease in value of capital goods over time*).

While GDP is the headline figure mainstream media usually follows, NDP is a more honest figure because it measures only the investment used for additional growth on the existing captial base, excluding the portion that goes into simply replacing worn down captial.

Core Concept 2.5: Calculating GDP (*Income Approach*)

The **income approach** calculates GDP by summing all incomes earned in production. This includes wages and salaries, interest income, rental income, profits (*including proprietor income and corporate profits*), and statistical adjustments (*depreciation, sales taxes, and net foreign factor income*). The equality of expenditure and income approaches reflects the circular flow of income.

$$\text{GDP} = \text{Wages} + \text{Interest} + \text{Rental Income} + \text{Profits} + \text{Statistical Adjustments}$$

Core Concept 2.6: Calculating GDP (Value Added Approach)

The **value added approach** calculates GDP by summing the value added at each stage of production. Value added equals the value of a firm's output minus the value of intermediate goods it purchases. This approach avoids double-counting by measuring only the incremental value contributed at each production stage.

$$\text{GDP} = \sum \text{Value Added at Each Stage}$$

For example, consider bread production: a farmer sells wheat for $1, a mill sells flour for $2, and a bakery sells bread for $3. The total value added is $1 + ($2 - $1) + ($3 - $2) = $3, which equals the final selling price.

The three methods of calculating GDP—expenditure, income, and value-added—are essentially three ways of counting the same dollar as it travels through the economy's circular flow. The expenditure approach catches that dollar at the cash register when a final good is purchased, the income approach tracks it as it is paid out to workers and business owners, and the value-added approach measures it as it is "built" piece-by-piece at every stage of production. Because every transaction is a closed loop where one person's spending is another person's income and a direct reflection of the new value created, all three paths inevitably lead to the exact same total: the market value of everything produced in a year—or GDP.

Core Concept 2.7: Limitations of GDP

Despite its widespread use, GDP has important limitations as a measure of economic well-being.

(a) GDP excludes **non-market transactions** and **underground economic activity**, such as household production, volunteer work, and illegal transactions.

(b) GDP does not capture changes in **environmental quality**, **income inequality**, or **social conditions**. A country with high GDP may also have significant pollution, income disparity, or social unrest.

(c) GDP does not account for **depreciation**, which overstates true economic welfare. NDP, as outlined in CORE CONCEPT REFERENCE, accounts for this inaccuracy.

Core Concept 2.8: Economic Growth

Economic Growth refers to a sustained increase in real GDP per capita (*per person*) over time, representing the expansion of an economy's productive capacity and living standards.

Long-run economic growth is represented by rightward shifts in the long-run aggregate supply curve (CORE CONCEPT REFERENCE) and outward shifts in the PPC (CORE CONCEPT REFERENCE). The aggregate production function expresses total output as a function of capital stock and human capital.

$$Y = A \cdot f(L, K, H, N)$$

Here,

(a) Y: Total Real Output (Real GDP).

(b) A: Total Factor Productivity (TFP). This represents technology, efficiency, and the "know-how" of the economy.

(c) L: Labor. The total number of workers or hours worked.

(d) K: Physical Capital. Tools, machinery, factories, and infrastructure.

(e) H: Human Capital. The skills, education, and experience of the workforce.

(f) N: Natural Resources. Land, minerals, and raw materials.

Labor productivity (*output per worker*) is a key determinant of economic growth. Factors driving growth include:

(a) **Capital accumulation**: Investment in machinery and infrastructure

(b) **Human capital development**: Education and skills training

(c) **Technological advancement**

(d) **Increases in the labor force participation rate** (CORE CONCEPT REFERENCE)

2.2 Labor Market

The **labor market** provides crucial information about employment, unemployment, and the economy's productive capacity.

Core Concept 2.9: Labor Force Participation Rate

The **labor force participation rate** (LFPR) measures the proportion of the working-age population actively in the labor market. It is calculated by the following equation.

$$\text{LFPR} = \left(\frac{\text{Labor Force}}{\text{Civilian Non-institutional Population}} \right) \times 100$$

The **labor force** is defined as the sum of all individuals who are either currently working or actively seeking work, excluding discouraged workers who have stopped seeking employment. The **civilian noninstitutional population** represents the "pool" individuals who are legally and physically able to participate in the economy.

Core Concept 2.10: Unemployment Rate

The **unemployment rate** (*or also called the U-3 rate*) measures the proportion of unemployed persons to the total labor force. This is a major headline statistic that all news outlets, economists, investors, etc. follow.

$$\text{Unemployment Rate} = \left(\frac{\text{Number of Unemployed}}{\text{Labor Force}} \right) \times 100$$

For many advanced economies, including the U.S., and unemployment rate of 3.5% to 4.5% is considered the "sweet spot" that governments strive for.

Core Concept 2.11: Types of Unemployment

Unemployment can be further categorized into the following baskets.

(a) **Frictional unemployment** occurs when workers are between jobs or searching for new employment. This type of unemployment is inevitable and even desirable in a dynamic economy, as it reflects workers seeking better matches.

(b) **Structural unemployment** arises from mismatches between worker skills and job requirements, often

due to technological change or shifts in the industrial composition of the economy. This unemployment persists even when the economy is at full employment.

(c) **Cyclical unemployment** results from economic recessions and fluctuations in aggregate demand. When the economy contracts, employers lay off workers, increasing cyclical unemployment.

The **natural rate of unemployment** (NRU) equals the sum of frictional and structural unemployment. It represents the unemployment rate when the economy is at full employment, consistent with stable inflation.

Core Concept 2.12: Underemployment

The **underemployment rate** (*also known as the U-6 rate*) represents the proportion of unemployed and underemployed persons to the total labor force. Underemployed persons consists of **marginally attached workers** (people who want a job and have looked for one in the last 12 months, but not in the last 4 weeks) and **part-time workers for economic reasons** (people who want full-time work (35+ hours) but are working part-time because their hours were cut or they cannot find a full-time position).

$$\text{Underemployment Rate} = \left(\frac{\text{Unemployed} + \text{Underemployed}}{\text{Labor Force} + \text{Marginally Attached}}\right) \times 100$$

Many economists argue that the U-6 rate provides a much more accurate picture of "labor underutilization" because it includes people who are technically working but are struggling.

2.3 Inflation

Core Concept 2.13: Inflation

Inflation represents a sustained increase in the general price level, equivalent to a decrease in the purchasing power of money.

Inflation is primarily kept track of by the following two statistics.

Core Concept 2.14: Consumer Price Index

The **Consumer Price Index (CPI)** is an inflationary measure which tracks the cost of a market basket of goods and services purchased by a typical consumer. This figure is a highly tracked statistic by new outlets and investors.

$$\text{CPI} = \frac{\text{Cost of market basket in current prices}}{\text{Cost of market basket in base period prices}} \times 100$$

This creates an index where the base period equals 100.

For the U.S., this figure is maintained by the Bureau of Labor Statistics (BLS) and is the main inflation figure used by the U.S. government for administrative and legal purposes.

Core Concept 2.15: Personal Consumption Expenditures

The **Personal Consumption Expenditures (PCE)** index is another inflation index that captures a broader range of consumer spending and accounts for substitution effects. This inflation measure is also another highly

tracked statistic.

For the U.S. this figure is maintained by Bureau of Economic Analysis and is the preferred inflation gauge for the Federal Reserve in the context of setting the Federal Reserve Rate (CORE CONCEPT REFERENCE).

Core Concept 2.16: Inflation Rate

The **inflation rate** equals the percentage change in the price level:

$$\text{Inflation Rate} = \frac{\text{Current Inflation Figure} - \text{Previous Inflation Figure}}{\text{Previous Inflation Figure}} \times 100$$

Economists and news outlets typically prioritize the **Year-over-Year (YoY)** rate because it provides a more stable picture of long-term trends by filtering out "noise" and seasonal fluctuations that often distort the **Month-over-Month (MoM)** data.

$$\text{MoM \%} = \left(\frac{\text{Value}_{currentmonth} - \text{Value}_{previousmonth}}{\text{Value}_{previousmonth}} \right) \times 100$$

$$\text{YoY \%} = \left(\frac{\text{Value}_{currentmonth} - \text{Value}_{samemonthlastyear}}{\text{Value}_{samemonthlastyear}} \right) \times 100$$

Core Concept 2.17: Nominal versus Real Variables

Nominal variables are measured in current dollars and are not adjusted for inflation. Examples include nominal GDP (*measured in current prices*) and nominal interest rates (*the stated rate without inflation adjustment*).

Real variables are adjusted for inflation and reflect actual purchasing power. Real GDP uses constant base-year prices. For instance, the following is the equation for **real interest rate**.

$$\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$$

However, most real interest rates reflect the expected future shift in inflation rather than realized inflation, thus the more accurate relationship is:

$$\text{Real Interest Rate} \approx \text{Nominal Interest Rate} - \text{Expected Inflation}$$

This distinction is crucial for economic decision-making, as borrowers and lenders care about real costs and returns. Lenders prefer to increase their lending rates when inflation is expected to rise, while borrowers prefer to delay borrowing when inflation is expected to fall. Consequently, inflation expectations serve as a focal point in determining the quantitative terms of debt contracts and financial arrangements. Many contracts, including loans, bonds, and wage agreements, are written in nominal terms, meaning their real value depends heavily on the inflation rate over their duration.

Unexpected inflation redistributes wealth from creditors to debtors. If a lender sets a nominal interest rate expecting low inflation but actual inflation turns out to be high, the real return becomes negative, benefiting the borrower at the lender's expense. This uncertainty complicates financial planning and investment decisions.

The **GDP deflator** provides another measure of the price level, representing how much of nominal GDP growth

is just due to inflation.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

2.4 Housing

Housing indicators provide valuable insights into the broader economic outlook. These statistics are closely watched because of housing's unique role in the economy.

Core Concept 2.18: Housing and Its Effects on the Economy

Although housing represents only approximately 3% of the economy, it serves as a powerful economic indicator for several reasons. Housing purchases typically involve 30-year mortgages, making housing finance deeply intertwined with the financial system. Additionally, housing purchases trigger purchases of complementary goods, including furniture, appliances, and home improvement materials.

The strong correlation between housing activity and GDP makes housing indicators valuable for predicting economic trends. The 2008 financial crisis demonstrated how housing market troubles can cascade through the entire financial system, particularly through mortgage-backed securities and derivatives tied to residential real estate.

Core Concept 2.19: Housing Starts

Housing starts (the number of new residential construction projects beginning) and existing home sales provide timely signals about consumer confidence and economic momentum.

2.5 Sentiment and Confidence Indicators

Core Concept 2.20: Business Confidence

Business confidence reflects firms' expectations about future demand and profitability.

Core Concept 2.21: Institute for Supply Management Purchasing Managers' Index

The Institute for Supply Management (ISM) publishes the Purchasing Managers' Index (PMI), the most widely followed indicator of U.S. manufacturing activity.

A PMI above 50 indicates expansion in the manufacturing sector and general economic optimism. A PMI below 50 indicates contraction and pessimism. The strong correlation between the PMI and GDP growth makes it a valuable leading indicator.

Core Concept 2.22: University of Michigan Consumer Sentiment Index

Consumer sentiment surveys, such as the University of Michigan Consumer Sentiment Index, measure household confidence about future economic conditions and their financial situations.

2.6 Economic Calendar and Indicator Releases

Economic data releases follow predictable schedules, allowing markets to anticipate and react to new information. Key releases include:

Inflation indicators: CPI (monthly), PCE (monthly), Producer Price Index (monthly), and inflation expectations surveys.

Labor market indicators: Unemployment rate (monthly), nonfarm payrolls (monthly), labor force participation rate, and various measures of underemployment (U-6).

Confidence indicators: PMI (monthly), consumer sentiment (preliminary and final releases).

Housing indicators: Housing starts, building permits, and existing/new home sales.

These indicators provide the foundation for macroeconomic analysis and policy formulation.

3 Macroeconomic Relationships

The macroeconomy functions through interconnected relationships between aggregate demand, aggregate supply, and various policy variables.

3.1 Aggregate Demand

Core Concept 3.1: Aggregate Demand

Aggregate Demand (AD) represents the total quantity of goods and services demanded in the economy at different price levels.

< PLOT OF AD CURVE >

The AD curve slopes downward for three reasons:

(a) **Wealth effect**: Lower price levels increase real wealth, allowing consumers to purchase more goods and services.

(b) **Interest rate effect**: Lower price levels reduce money demand, causing interest rates to fall, stimulating investment.

(c) **Exchange rate effect**: Lower interest rates cause currency depreciation, making domestic goods cheaper abroad and stimulating exports.

Rightwards Shifts in the AD curve result from changes in the following four components. Correspondingly, decreases shift the AD leftward.

(a) **Consumer spending**: Increases from tax cuts or increased confidence shift AD rightward.

(b) **Investment**: Increases from business optimism or technological innovation shift AD rightward.

(c) **Government spending**: Increases shift AD rightward.

(d) **Net exports**: Increases from foreign economic growth or currency depreciation shift AD rightward.

Core Concept 3.2: The Multiplier Effect

An initial change in spending creates a larger final change in GDP through the multiplier effect. To understand why this is, we introduce two new concepts: the **Marginal Propensity to Consume or MPC** (the fraction of additional disposable income that households spend) and the **Marginal Propensity to Save (MPS)** (the fraction of additional household income that housholds save). Thus,

$$\text{MPC} + \text{MPS} = 1$$

The **spending multiplier** effect operates through successive rounds of spending. When one person spends income, the recipient receives income and spends a portion (*the MPC*), continuing until leakages (*saving, taxes, imports*) absorb the remaining income.

$$\text{Spending Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

The **tax multiplier** works in the same manner but with a smaller initial impact because a tax change first affects disposable income rather than direct aggregate demand. When taxes are cut, consumers do not spend the entire amount; they save a portion (*the MPS*) immediately. Consequently, the first round of the spending cycle is "diluted" by this initial leakage, making the tax multiplier's overall effect less powerful than that of the spending multiplier. The negative sign in the following equation for the tax multiplier reflects that tax increases reduce disposable income and spending.

$$\text{Tax Multiplier} = -\frac{\text{MPC}}{1 - \text{MPC}}$$

3.2 Aggregate Supply

Core Concept 3.3: Short-Run Aggregate Supply

Short-Run Aggregate Supply (SRAS) represents the total output firms are willing to produce at different price levels in the short run, when some input costs (*particularly wages*) are fixed.

< PLOT OF SRAS >

The SRAS curve slopes upward because higher price levels increase profitability when input costs are **sticky**. Firms respond to higher prices by increasing production to capture additional profits in the near future until productions cost adjust.

The SRAS curve illustrates the short-run trade-off between inflation and unemployment. Points higher on the curve indicate economic boom conditions (*low unemployment, high inflation*), while points lower on the curve indicate recessionary conditions (*high unemployment, low inflation*).

SRAS shifts result from changes in productivity, input costs, labor market conditions, and taxes or subsidies.

Core Concept 3.4: Long-Run Aggregate Supply

Long-Run Aggregate Supply (LRAS) represents the economy's potential output when all input costs can fully adjust to price changes. In the long run, output is independent of the price level and equals potential GDP.

< PLOT OF LRAS >

3.3 Economic Theory

Core Concept 3.5: Types of Inflation

Demand-pull inflation results from AD increasing beyond potential output, pushing the economy above its LRAS.

< PLOT OF DEMAND PULL INFLATION >

Cost-push inflation results from SRAS decreasing due to supply shocks, reducing output and raising prices simultaneously.

< PLOT OF COST PUSH INFLATION >

Core Concept 3.6: Long-Run Self-Adjustment & Equilibrium

Without government intervention, the economy naturally adjusts to long-run equilibrium through wage and price flexibility. When output falls below potential (*recessionary gap*), unemployment rises, wages decline, and production costs fall, shifting SRAS rightward until output returns to potential. When output exceeds potential (*inflationary gap*), upward pressure on wages increases costs, shifting SRAS leftward.

< PLOT OF SELF ADJUSTMENT ALONG THE AD, SRAS, and LRAS >

This self-adjustment process explains why classical economists emphasized the economy's natural tendency toward full employment (U-3 Rate = NRU), though the adjustment process may be slow and painful.

Core Concept 3.7: The Phillips Curve

The **Phillips Curve** illustrates the short-run inverse relationship between unemployment and inflation. Lower unemployment corresponds to higher inflation, reflecting the short-run trade-off between economic activity and price stability.

< PLOT OF PHILLIPS CURVE >

Movements along the SRAS correspond to movements along the Short-Run Phillips Curve (SRPC). Rightward shifts in SRAS shift the SRPC leftward (better outcomes for both unemployment and inflation).

In the long run, the economy returns to the natural rate of unemployment regardless of inflation, making the Long-Run Phillips Curve vertical.

Core Concept 3.8: Quantity Theory of Money

The **Quantity Theory of Money** provides a mathematical framework for understanding the long-run causes of inflation. The theory is expressed through the following identity.

$$MV = PQ$$

Here, M represents the **money supply**, V represents **velocity** (*how frequently a unit of money changes hands*), P represents the **price level**, and Q represents the **quantity of real output** (*or real GDP*).

Classical economists assume that both velocity (V) and real output (Q) are relatively stable or determined by non-monetary factors in the long run.

- **Velocity** is determined by spending habits and banking technology.

- **Output** is determined by the economy's productive capacity (*labor, capital, and technology*).

Because V and Q are assumed to be stable, any change in the money supply (M) must result in a proportional change in the price level (P). Therefore, sustained inflation—a persistent rise in P—is interpreted as the direct result of a sustained increase in the money supply.

4 Financial Sector

The financial sector channels savings to investment and facilitates economic transactions.

4.1 Money and the Money Market

Core Concept 4.1: Functions of Money

Money serves the following three essential functions in an economy.

- **Medium of exchange**: Money facilitates transactions by eliminating barter's double coincidence of wants.
- **Store of value**: Money preserves purchasing power for future use.
- **Unit of account**: Money provides a common measure for valuing goods and services.

When hyperinflation erodes money's value, it fails as a store of value and unit of account, disrupting economic activity.

The **opportunity cost of holding money** is the interest that could be earned by holding interest-bearing assets instead. Higher interest rates increase this opportunity cost, reducing money demand because people would much rather earn interest from securities.

Core Concept 4.2: Money Supply (M0, M1, and M2)

Economists categorize the money supply using the following aggregates.

- **M0**: Currency in circulation plus reserves held by banks at the central bank.
- **M1**: The most liquid forms, including currency, checking accounts, and travelers' checks.
- **M2**: Includes M1 plus savings deposits, small time deposits, and money market mutual funds.

< DIAGRAM OF MONEY SUPPLY >

These aggregates reflect different degrees of liquidity, with broader measures (*M2*) capturing more forms of money that can be relatively easily converted to cash.

Core Concept 4.3: The Money Market

The money market determines the equilibrium nominal interest rate through the interaction of money supply and money demand. The money supply curve is vertical (*set by the central bank*), while money demand slopes downward (*higher interest rates increase the opportunity cost of holding money*).

< PLOT OF MONEY MARKET >

Factors affecting money demand include price levels (*higher prices require more money for transactions*), national income (*higher income increases transactions*), and technology affecting liquidity.

4.2 Banking System

Core Concept 4.4: Fractional Reserve Banking

Banks operate under **fractional reserve banking**, maintaining only a fraction of deposits as **reserves** while lending the remainder. This reserve to lending ratio is set by the **required reserve ratio (RRR)**, which establishes the minimum fraction of deposits that banks must hold as reserves.

This system creates money through the multiplier effect as shown by the following formulas.

$$\text{money multiplier} = \frac{1}{\text{RRR}}$$

$$\Delta\text{Money Supply} = \text{Excess Reserves} \times \frac{1}{rr}$$

As these loans are spent and redeposited into other banks, the process repeats across the banking system, mathematically expanding a single initial deposit into a much larger total increase in the money supply. We have seen this same pattern with the multiplier effect in the context of MPC and MPS (CORE CONCEPT REFERENCE). With a 20% RRR, an initial deposit of $1,000 can ultimately support $5,000 in total deposits.

Core Concept 4.5: Loanable Funds Supply and Demand

The loanable funds market determines the real interest rate through the supply of funds (*savings*) and demand for funds (*investment*). With higher real interest rates, the supply of loanable funds increases (*encouraging saving*) and the demand decreases (*discouraging borrowing for investment*).

< PLOT OF LOANABLE FUNDS GRAPH >

The position of these curves reflects national saving behavior, government borrowing, and investment profitability expectations.

Core Concept 4.6: National Savings

National Savings represents total saving in an economy for investment, equaling private savings plus public savings. A **closed economy** national savings is represented by the following equation.

$$I = \text{National Savings} = Y - C - G = \underbrace{(Y - T - C)}_{\text{Private Saving}} + \underbrace{(T - G)}_{\text{Public Saving}}$$

As the above equations shows nicely, private saving is simply houshold income (Y) after taxes (T) and private consumption (C) and public saving (*or the government saving*) its tax revenue (T) after accounting for its spending (G).

In an **open economy**, the relationship between savings and investment changes because a country is no longer limited to its own internal resources. It can interact with the the rest of the world and must consider net capital outflow (NCO).

$$\text{National Savings} = I + NCO$$

Net Capital Outflow (NCO) measures the net flow of funds being invested abroad, calculated by the following equation.

$$\text{NCO} = \text{Domestic purchase of foreign assets} - \text{Foreign purchase of domestic assets}$$

5 Stabilization Policies

Policymakers use fiscal and monetary policy to manage aggregate demand and stabilize the economy.

5.1 Fiscal Policy

Core Concept 5.1: Fiscal Policy

Fiscal policy involves government spending and taxation decisions to influence aggregate demand.

- **Contractionary fiscal policy** (*reducing spending or increasing taxes*) reduces AD to combat inflation.
- **Expansionary fiscal policy** (*increasing spending or decreasing taxes*) stimulates AD to combat recession.

Keep in mind that the initial impact of fiscal policy is multiplied through the multiplier effect (CORE CONCEPT REFERENCE), meaning the ultimate change in GDP exceeds the initial fiscal impulse.

Core Concept 5.2: Automatic Stabilizers

Automatic stabilizers are fiscal policies that automatically counteract economic fluctuations without discretionary action by the government. These stabilizers provide immediate response to economic conditions, avoiding recognition and implementation lags that affect discretionary policy.

Progressive income taxes exemplify this: during expansions, rising incomes push taxpayers into higher brackets, automatically increasing tax revenues and exerting downward pressure on the economy. During recessions, falling incomes reduce tax burdens, providing automatic stimulus.

Core Concept 5.3: Crowding Out and Government Debt

Government **deficit** (*annual spending minus revenue*) and **debt** (*accumulated deficits*) represent the fiscal stance. Deficit spending can stimulate the economy but may crowd out private investment through higher interest rates.

The **crowding out effect** occurs when government borrowing increases demand for loanable funds, raising interest rates and reducing private investment. This effect is more significant in closed economies with limited capital inflow.

5.2 Monetary Policy

Core Concept 5.4: Monetary Policy

Monetary policy involves central bank actions controlling the money supply and interest rates.

- **Expansionary monetary policy** decreases interest rates and increases the money supply to stimulate economic activity.
- **Contractionary monetary policy** increases interest rates and reduces the money supply to combat inflation.

The central bank influences rates through open market operations, reserve requirements, and the discount rate.

The central bank influences rates through the following monetary policy tools.

Core Concept 5.5: Monetary Policy Tool (*Open Market Operations*)

Open market operations involve buying and government bonds to affect bank reserves.

- **Purchasing bonds** injects reserves and expand the money supply in the economy
- **Selling bonds** build up reserves and contract the money supply in the economy

Core Concept 5.6: Monetary Policy Tool (*Required Reserve Ratio*)

Changes in the **required reserve ratio (RRR)** affect banks' lending capacity. Lowering the RRR expands the money supply and raising it contracts the money supply. Keep in mind that the ultimate change in money supply is amplified due to the multiplier effect in fractional reserve banking (CORE CONCEPT REFERENCE).

Historically, the RRR was set by the Federal Reserve and publicly announced. However, as of March 2020 (*COVID pandemic*), the Fed reduced the RRR to 0% for all deposits, effectively eliminating it as an active policy tool (LINK TO WEBSITE - https://www.federalreserve.gov/monetarypolicy/reservereq.htm).

In contrast, some central banks (*such as the People's Bank of China*) continue to actively adjust reserve requirements as a monetary policy tool.

Core Concept 5.7: Monetary Policy Tool (*Discount Rate, Interest on Reserves, and Federal Funds Rate*)

The **discount rate** is the interest rate banks pay to borrow from the central bank. Consequently, no bank would ever rationally borrow from another commercial bank with a higher interest rate than the discount rate. Thus, lowering the discount rate encourages borrowing because it effectively acts as the ceiling for all other interest rates; raising it has the opposite effect.

Interest on reserves (IOR) is the rate of interest that a central bank pays to commercial banks on the funds they hold in their reserve accounts at the central bank. This effectively sets a floor for interest rates because no lender would ever rationally lend at a rate below the IOR.

Through lowering and raising these two figures, the central bank can effectively manage short-term rates across the economy, influencing the level of economic activity as a result. This target rate range is called the **Federal Reserve Rate (FFR)**.

< PLOT OF FFR >

*The FFR serves as the primary benchmark for the global financial system, effectively anchoring the interest rates for all other types of borrowing. To maintain economic stability, the **Federal Open Market Committee (FOMC)** meets eight times annually to evaluate and adjust the FFR target.*

The Federal Reserve operates under a dual mandate, seeking to maintain an economy characterized by price stability—quantified as a 2% long-run inflation target—and maximum employment, which is traditionally associated with a NRU centered around 4%.

6 International Trade

6.1 Balance of Payments

The **Balance of Payments (BOP)** is a comprehensive record of all economic transactions between the residents of a country and the rest of the world over a specific period. It is composed of two sections.

Core Concept 6.1: Current Account

The **current account** measures the net balance of **value** into a country in the form of goods, services, income, and transfers (*cross border value flows*). It essentially represents a nation's short-term economic transactions.

A current account deficit means a country is importing more than it exports and is a net borrower from abroad. A surplus indicates the opposite.

It's helpful to think of the current account as a country's shopping bag.

Core Concept 6.2: Capital (Financial) Account

The **capital (financial) account** records the net change in national ownership of assets. It reflects how a country finances the items in its current account or where it invests its current account surplus.

It's helpful to think of the capital account as a country's wallet.

In theory, the current account and capital account sum to zero: a current account deficit must be financed by a capital account surplus (net borrowing from abroad).

6.2 Exchange Rates

Core Concept 6.3: Exchange Rates

The **exchange rate** represents the price of one currency in terms of another. Currency **appreciation** means a currency becomes more valuable relative to another, and **depreciation** means it becomes less valuable.

< PLOT OF EXCHANGE RATE GRAPH >

Like other markets, exchange rates are determined by supply and demand. Factors affecting currency demand include the following.

- relative interest rates (*higher rates attract investors towards assets that are bought via that currency*)
- relative price levels (*lower prices increase global economy competitiveness*)
- relative national income (*higher income increases imports and currency demand*)
- tariffs (*tariffs reduce import demand*).

Core Concept 6.4: Real Interest Rate Parity

The **Real Interest Rate Parity** condition states that differences in real interest rates between countries drive capital flows. Higher domestic real interest rates relative to foreign rates attract capital inflows, appreciating the domestic currency.

These capital flows continue until real interest rates equalize across countries, implying that in integrated capital markets, real interest rates should converge globally.

In practice, real interest rates do not perfectly converge due to the following. Consequently, the RIRP condition serves rather as a long-run tedency than a constant state of equilibrium.

- **Transaction Costs:** Fees and regulations that impede the free flow of capital.
- **Risk Premiums:** Higher rates required to compensate for political or default risk in specific countries.

- **Non-Tradable Goods:** Differences in local inflation that prevent Purchasing Power Parity (PPP) from holding.